Titouan Verdier

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Profile

Telecommunications Engineering student at INSA Lyon and clarinetist, combining technical rigor, creativity, and curiosity. Passionate about software development, I enjoy tackling new challenges and continuously learning.

Education

INSA Lyon *Sept 2022 – Aug 2027*
Telecommunications Engineering - Networks, Software Development, Signal Processing

Sogang University - 서강대학교 (Academic Exchange) *Sept 2026 – Dec 2026*
Computer Science & Electrical Engineering

CRC Saint-Priest
Clarinet (Music Studies Certificate - CEM), Music Theory, Concert Band

Experience

Geographic Information Systems (GIS) Software Developer Intern *Lyon, France*
Worldline, Magellan *Apr 2026 – Aug 2026*

- Development of a web application for filing and processing grant applications related to the CAP (Common Agricultural Policy).

Equipment Manager *Lyon, France*
INSA Music Association (AMI) *Jul 2024 – Jun 2025*

- Managed the music association's equipment inventory and maintenance.
- Organized and tracked equipment loans to members and other student associations at INSA.

Projects

Home Automation Project – *Home Assistant, Networking, Zigbee, MQTT* *2026*

- Designed and deployed a comprehensive home automation infrastructure (temperature/humidity sensors, automation scenarios, lighting, dashboards, security cameras, remote access, alarm system).

FPGA Audio Dome – *Faust, FPGA Board* *2025*

- Research project focused on using FPGA boards for 3D audio spatialization within a dome setup.
- Integration as a plugin for Digital Audio Workstations (DAW).

SON Project – *C++, Faust, Arduino, Teensy* *github.com/m-nsx/son*

- Developed a polyphonic digital synthesizer with a dedicated physical hardware interface.

WEB Project – *MongoDB, Express, React, Node.js* *github.com/m-nsx/swu-webapp*

- Developed a web application to manage SWU card collections (MERN Stack).
- Implemented an automatic data synchronization module via an external API.

NAS & Game Server Project – *TrueNAS, Networking, Docker* *webmap*

- Deployed and secured servers running TrueNAS and Ubuntu on repurposed hardware, featuring secure remote access.

INSPIRE Project – *Project Management, Planning, Pedagogy* *2025*

- Hosted an open-access workshop at INSA on building production-ready NAS systems (security, reliability, availability, and remote access).
- Authored and published a guide addressing personal data sovereignty challenges.

Skills

Languages: French (Native), English (B2-C1), German (A2), Japanese (Beginner)

Languages & Technologies: Python, HTML/CSS, JavaScript, C++, MERN Stack, Computer Networks, Git, Docker, CAD, DAW / Music Production